EXP.NO: 7	Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits.
DATE:	

AIM:

To create a Java Servlet that stores user preferences such as theme and language using cookies, and retrieves and displays these preferences when the user revisits the application.

ALGORITHM:

1. Input:

- The user selects preferences such as:
  - Theme (e.g., Light/Dark)
  - Language (e.g., English/Tamil)
- The form submits this data to the servlet.

2. Steps for Creating and Handling Cookies in Servlet:

1 . Create an HTML Form:

- Design an HTML form with:
  - Dropdown for selecting theme
  - Dropdown for selecting language
- Add a submit button
- Set the action attribute to the servlet URL
- Use the POST method

2. Create the Servlet to Handle Preferences:

- Write a Java Servlet class
- Override the doPost() and doGet() methods

3. Store Preferences Using Cookies (doPost method):

- Retrieve user inputs using:

- Create cookies:
- Set cookie expiry time:
- Add cookies to response:

4. Retrieve Preferences Using Cookies (doGet method):

- Get cookies from request:
- Loop through cookies and extract values:

5. Display Preferences:

- Generate HTML response showing:
    - Selected theme
    - Selected language
- If cookies exist → display saved preferences
- If not → display default message

6. Deploy and Test the Servlet:

- Compile servlet
- Deploy in a server (e.g., Apache Tomcat)
- Open browser and submit preferences • Refresh or revisit to check stored values

7. Observe Results:

- First visit → preferences are stored in cookies
- Subsequent visits → preferences are retrieved automatically

index.html:

```
<!DOCTYPE html>
<html>
<head>
   <title>Online Voting Preferences</title>
   <style>
```

```
        body {
            background-color: #f3e5f5;
            text-align: center;
            margin-top: 100px;
            font-family: Arial;
        }
        form {
            background: white;
            padding: 20px;
            display: inline-block;
            border-radius: 10px;
        }
        select, button {
            margin: 10px;
            padding: 8px;
        }
        button {
            background: #9c27b0;
            color: white;
            border: none;
            border-radius: 5px;
        }
    </style>
</head>
<body>
```

```
<h2>Online Voting - User Preferences</h2>
<form action="PreferenceServlet" method="post">
   <label>Choose Theme:</label><br>
   <select name="theme">
      <option value="Light">Light</option>
      <option value="Dark">Dark</option>
   </select><br>
   <label>Choose Language:</label><br>
   <select name="language">
      <option value="English">English</option>
      <option value="Tamil">Tamil</option>
   </select><br>
   <button type="submit">Save Preferences</button>
</form>
<br><br>
<a href="PreferenceServlet">View Saved Preferences</a>
</body>
</html>
```

PreferenceServlet.java:

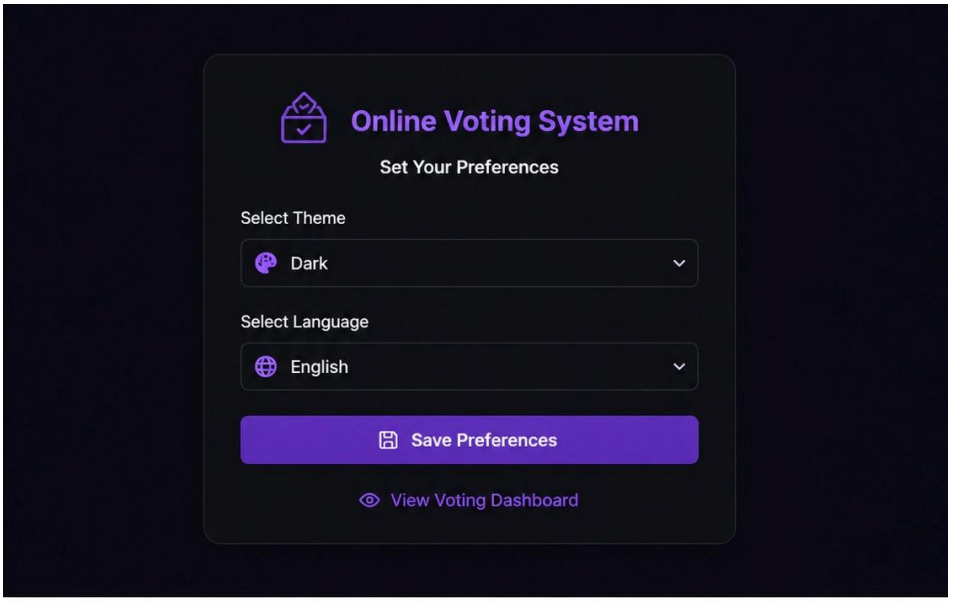
```
package com.example.voting;
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
```

```
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/PreferenceServlet")
public class PreferenceServlet extends HttpServlet {
    // Save preferences (POST)
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        String theme = request.getParameter("theme");
        String language = request.getParameter("language");        // Create cookies
        Cookie themeCookie = new Cookie("theme", theme);
        Cookie langCookie = new Cookie("language", language);
        // Set cookie expiry (1 day)
        themeCookie.setMaxAge(24 * 60 * 60);
        langCookie.setMaxAge(24 * 60 * 60);
        // Add cookies to response
        response.addCookie(themeCookie);
        response.addCookie(langCookie);
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<html><body style='text-align:center;
background:#e1bee7;'>");
        out.println("<h2>Preferences Saved Successfully</h2>");
        out.println("<a href='PreferenceServlet'>View Preferences</a>");
        out.println("</body></html>");
    }
```

```
    // Retrieve preferences (GET)
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        Cookie[] cookies = request.getCookies();
        String theme = "Not Set";
        String language = "Not Set";
       if (cookies != null) {
            for (Cookie c : cookies) {
                if (c.getName().equals("theme")) {
                    theme = c.getValue();
                }
                if (c.getName().equals("language")) {
                    language = c.getValue();
                }
            }
        }
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<html><body style='text-align:center;
background:#d1c4e9;'>");
        out.println("<h2>User Preferences</h2>");
        out.println("<p><b>Theme:</b> " + theme + "</p>");
        out.println("<p><b>Language:</b> " + language + "</p>");
        out.println("<a href='index.html'>Go Back</a>");
```

```
        out.println("</body></html>");
    }
}
```

OUTPUT:



Result:

A Servlet program is successfully created to store user preferences using cookies and retrieve them on subsequent visits.